

Youyi Yan

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SUMMARY

Computer Science student at the **University of Waterloo**. Experienced in **Go, SQL, C++, TypeScript, React, networking, backend architecture, and CI/CD**, with hands-on experience building backend services, authentication systems, databases, and hardware integrations.

WORK EXPERIENCE

Software Developer Intern – Radox Health, Antrim, UK

Jan 2026 - Aug 2026

- Served as the **primary developer** of a production **Laboratory Information Management System (LIMS)**, building backend, database, and frontend systems using **Go, SQL Server, React, and TypeScript**; the platform has been publicly released, deployed internally, and received confirmed purchase interest from external companies.
- Implemented persistent **TCP/IP analyser integrations**, including socket listeners, **ASTM/HL7** protocol parsing, and **asynchronous workers** for validating and processing incoming results.
- Built a centralized **OAuth 2.0-based SSO service** adopted as the standard login system across Radox internal applications, with RBAC, application-level authorization, and auditing.
- Integrated **Azure DevOps CI/CD pipelines** for automated builds, deployment, and health checks.

Software Developer – NB INC dba STEMBlock, NB, CA

May 2025 - Aug 2025

- Built a student portal backend in **Go** with **Gin, GORM, and PostgreSQL**, implementing authentication, rate limiting, sessions, and RBAC; deployed on **DigitalOcean** with automated CI/CD.
- Designed and taught an introductory **Python** course to children with hands-on programming exercises.

TECHNICAL SKILLS

Languages:	Go, Python, C, C++, C#, Kotlin, SQL, TypeScript, HTML
Frameworks & Databases:	Gin, GORM, React, Next.js, Vite, Flask, SQL Server, PostgreSQL
Systems & Tools:	TCP/IP, REST APIs, OAuth 2.0, Azure, Docker, Git, Linux, Godot

PROJECTS

WaterlooStar.com – Student Community Forum

waterloostar.com

- Founded and leading development of a University of Waterloo community platform, with a deployed prototype for housing, campus events, and course resources. Designed the backend architecture and REST APIs in **Go** with **PostgreSQL**, with automated deployment to cloud infrastructure.

Aetheria – 2D Turn-Based Roguelike

- Independently developing **Aetheria**, a 2D turn-based roguelike in **C#/.NET** with **Godot**, featuring a finite state machine turn loop, procedural generation, enemy AI, and a data-driven layer built on **interfaces, polymorphism, and event-driven** design for unit testing.

Spinema – Group Movie Selection App

- Developed an **Android** app in **Kotlin** with **Android Studio** for group movie selection, implemented room creation, voting, weighted selection, and the supporting application architecture and UI flows.

EDUCATION

2023 - Present Bachelor of Computer Science, **University of Waterloo**

Expected 2028